

Лабораторная работа № 14. Статическая маршрутизация в Интернете. Настройка

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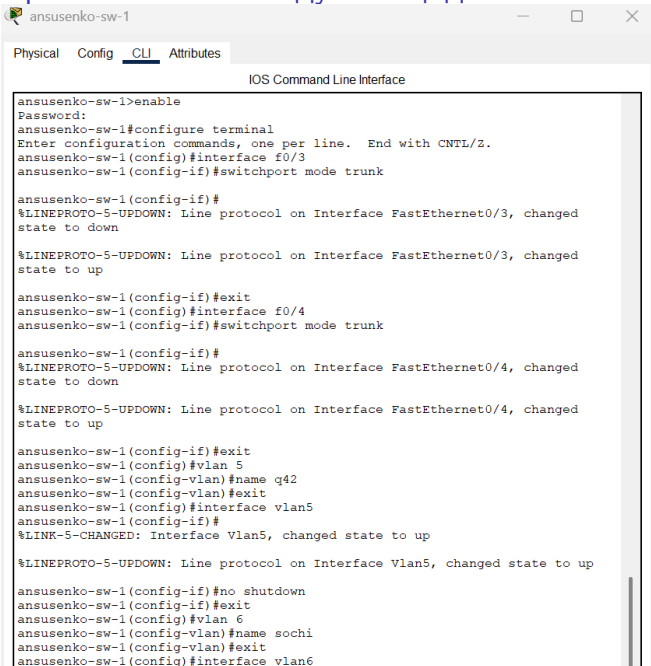
Российский университет дружбы народов имени Патриса Лумумбы

2026

Цель работы

- ▶ Настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.

Настройка связи между площадками



```
ansusenko-sw-1>enable
Password:
ansusenko-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ansusenko-sw-1(config)#interface f0/3
ansusenko-sw-1(config-if)#switchport mode trunk

ansusenko-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed
state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed
state to up

ansusenko-sw-1(config-if)#exit
ansusenko-sw-1(config)#interface f0/4
ansusenko-sw-1(config-if)#switchport mode trunk

ansusenko-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed
state to down

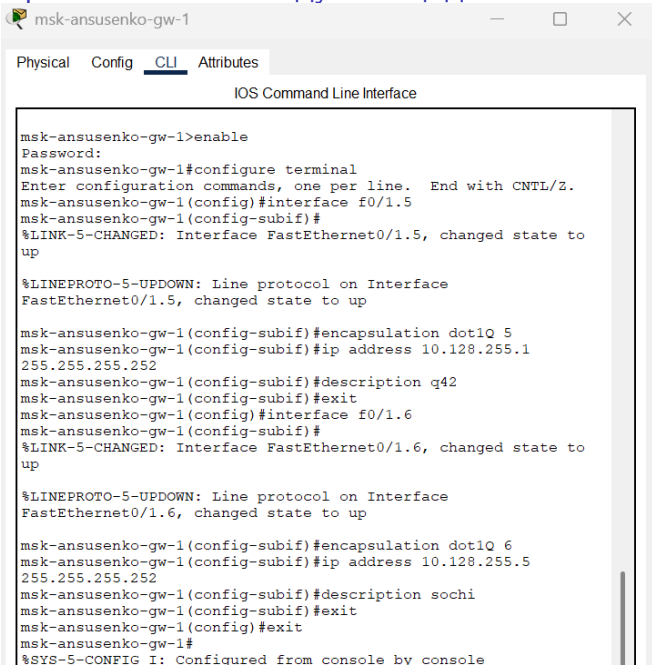
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed
state to up

ansusenko-sw-1(config-if)#exit
ansusenko-sw-1(config)#vlan 5
ansusenko-sw-1(config-vlan)#name q42
ansusenko-sw-1(config-vlan)#exit
ansusenko-sw-1(config)#interface vlan5
ansusenko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up

ansusenko-sw-1(config-if)#no shutdown
ansusenko-sw-1(config-if)#exit
ansusenko-sw-1(config)#vlan 6
ansusenko-sw-1(config-vlan)#name sochi
ansusenko-sw-1(config-vlan)#exit
ansusenko-sw-1(config)#interface vlan6
```

Настройка связи между площадками



The screenshot shows a network configuration window for a device named 'mks-ansusenko-gw-1'. The window has tabs for 'Physical', 'Config', 'CLI' (which is selected), and 'Attributes'. The main area is titled 'IOS Command Line Interface' and displays a series of configuration commands and their outputs. The configuration includes enabling the device, setting a password, configuring interfaces f0/1.5 and f0/1.6 with encapsulation, IP addresses, and descriptions. The interface f0/1.5 is configured with IP 10.128.255.1/24 and description 'q42'. The interface f0/1.6 is configured with IP 10.128.255.5/24 and description 'sochi'. The configuration is saved to the startup configuration.

```
mks-ansusenko-gw-1>enable
Password:
mks-ansusenko-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
mks-ansusenko-gw-1(config)#interface f0/1.5
mks-ansusenko-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.5, changed state to
up

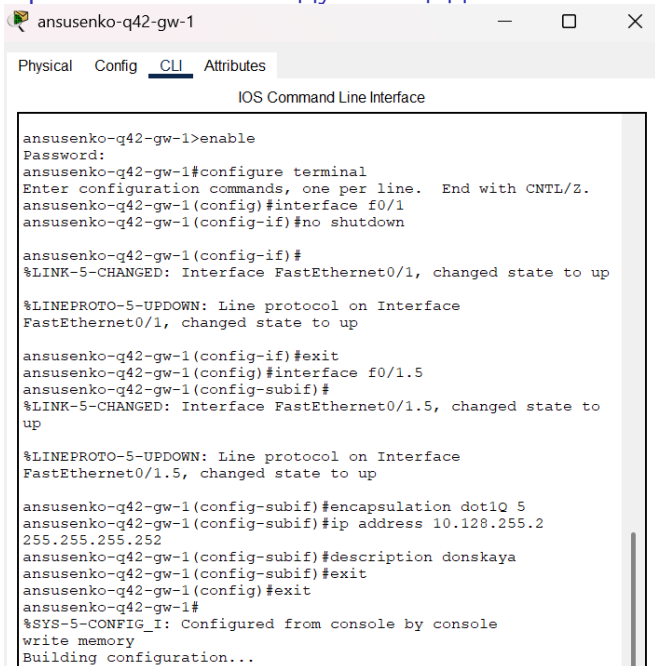
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1.5, changed state to up

mks-ansusenko-gw-1(config-subif)#encapsulation dot1Q 5
mks-ansusenko-gw-1(config-subif)#ip address 10.128.255.1
255.255.255.252
mks-ansusenko-gw-1(config-subif)#description q42
mks-ansusenko-gw-1(config-subif)#exit
mks-ansusenko-gw-1(config)#interface f0/1.6
mks-ansusenko-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.6, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1.6, changed state to up

mks-ansusenko-gw-1(config-subif)#encapsulation dot1Q 6
mks-ansusenko-gw-1(config-subif)#ip address 10.128.255.5
255.255.255.252
mks-ansusenko-gw-1(config-subif)#description sochi
mks-ansusenko-gw-1(config-subif)#exit
mks-ansusenko-gw-1(config)#exit
mks-ansusenko-gw-1#
%SYS-5-CONFIG I: Configured from console by console
```

Настройка связи между площадками



```
ansusenko-q42-gw-1
Physical  Config  CLI  Attributes
IOS Command Line Interface

ansusenko-q42-gw-1>enable
Password:
ansusenko-q42-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ansusenko-q42-gw-1(config)#interface f0/1
ansusenko-q42-gw-1(config-if)#no shutdown

ansusenko-q42-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

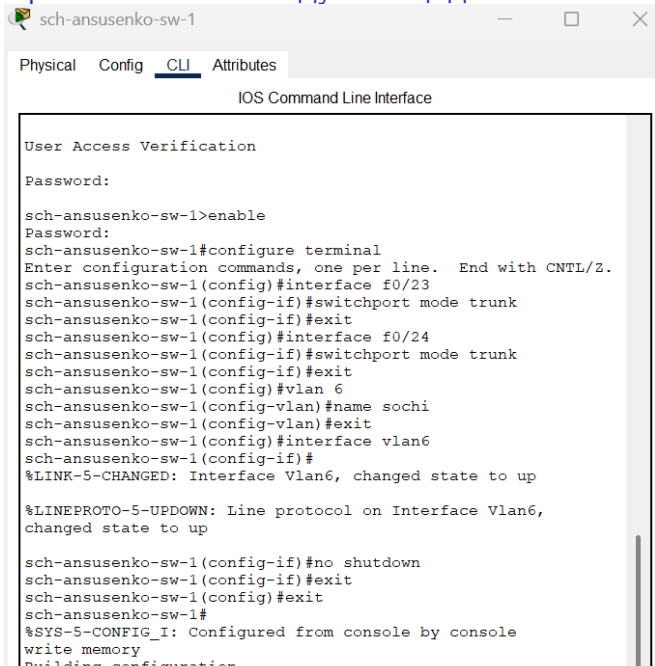
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to up

ansusenko-q42-gw-1(config-if)#exit
ansusenko-q42-gw-1(config)#interface f0/1.5
ansusenko-q42-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.5, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1.5, changed state to up

ansusenko-q42-gw-1(config-subif)#encapsulation dot1Q 5
ansusenko-q42-gw-1(config-subif)#ip address 10.128.255.2
255.255.255.252
ansusenko-q42-gw-1(config-subif)#description donskaya
ansusenko-q42-gw-1(config-subif)#exit
ansusenko-q42-gw-1(config)#exit
ansusenko-q42-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
```

Настройка связи между площадками



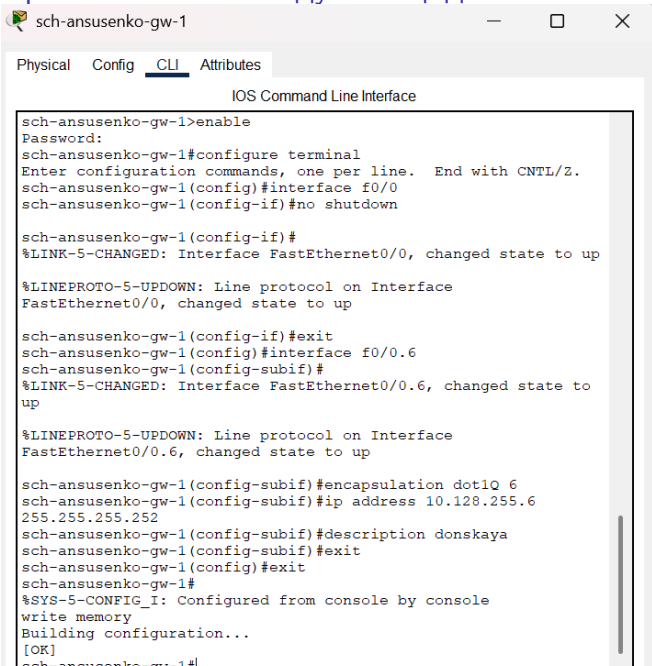
The screenshot shows a terminal window titled 'sch-ansusenko-sw-1' with tabs for 'Physical', 'Config', 'CLI' (selected), and 'Attributes'. The main title is 'IOS Command Line Interface'. The terminal output shows the following sequence of commands and responses:

```
User Access Verification
Password:
sch-ansusenko-sw-1>enable
Password:
sch-ansusenko-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-ansusenko-sw-1(config)#interface f0/23
sch-ansusenko-sw-1(config-if)#switchport mode trunk
sch-ansusenko-sw-1(config-if)#exit
sch-ansusenko-sw-1(config)#interface f0/24
sch-ansusenko-sw-1(config-if)#switchport mode trunk
sch-ansusenko-sw-1(config-if)#exit
sch-ansusenko-sw-1(config)#vlan 6
sch-ansusenko-sw-1(config-vlan)#name sochi
sch-ansusenko-sw-1(config-vlan)#exit
sch-ansusenko-sw-1(config)#interface vlan6
sch-ansusenko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6,
changed state to up

sch-ansusenko-sw-1(config-if)#no shutdown
sch-ansusenko-sw-1(config-if)#exit
sch-ansusenko-sw-1(config)#exit
sch-ansusenko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
```

Настройка связи между площадками



The screenshot shows a network configuration window titled "sch-ansusenko-gw-1". It has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The window displays the "IOS Command Line Interface". The following commands and their outputs are shown:

```
sch-ansusenko-gw-1>enable
Password:
sch-ansusenko-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
sch-ansusenko-gw-1(config)#interface f0/0
sch-ansusenko-gw-1(config-if)#no shutdown

sch-ansusenko-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

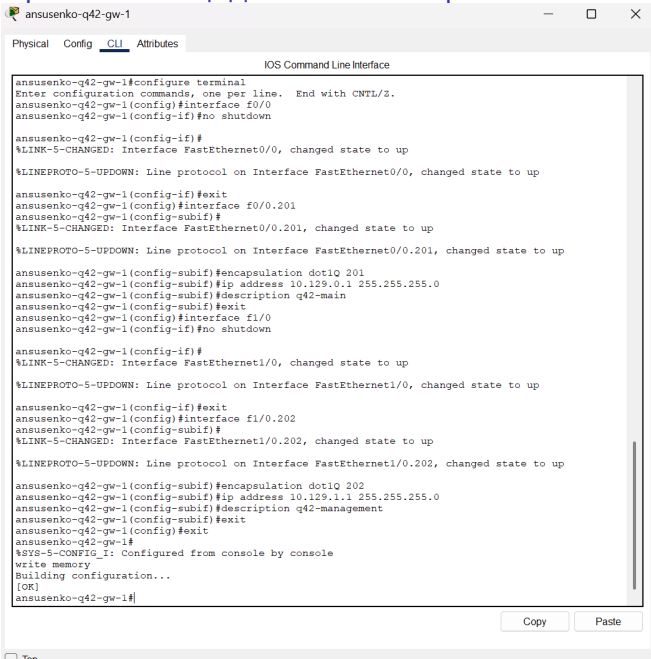
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up

sch-ansusenko-gw-1(config-if)#exit
sch-ansusenko-gw-1(config)#interface f0/0.6
sch-ansusenko-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.6, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0.6, changed state to up

sch-ansusenko-gw-1(config-subif)#encapsulation dot1Q 6
sch-ansusenko-gw-1(config-subif)#ip address 10.128.255.6
255.255.255.252
sch-ansusenko-gw-1(config-subif)#description donskaya
sch-ansusenko-gw-1(config-subif)#exit
sch-ansusenko-gw-1(config)#exit
sch-ansusenko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
sch-ansusenko-gw-1#
```

Настройка площадки 42-го квартала



```
ansusenko-q42-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ansusenko-q42-gw-1(config)#interface f0/0
ansusenko-q42-gw-1(config-if)#no shutdown

ansusenko-q42-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

ansusenko-q42-gw-1(config-if)#exit
ansusenko-q42-gw-1(config)#interface f0/0.201
ansusenko-q42-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up

ansusenko-q42-gw-1(config-subif)#encapsulation dot1Q 201
ansusenko-q42-gw-1(config-subif)#ip address 10.129.0.1 255.255.255.0
ansusenko-q42-gw-1(config-subif)#description q42-main
ansusenko-q42-gw-1(config-subif)#exit
ansusenko-q42-gw-1(config)#interface f1/0
ansusenko-q42-gw-1(config-if)#no shutdown

ansusenko-q42-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

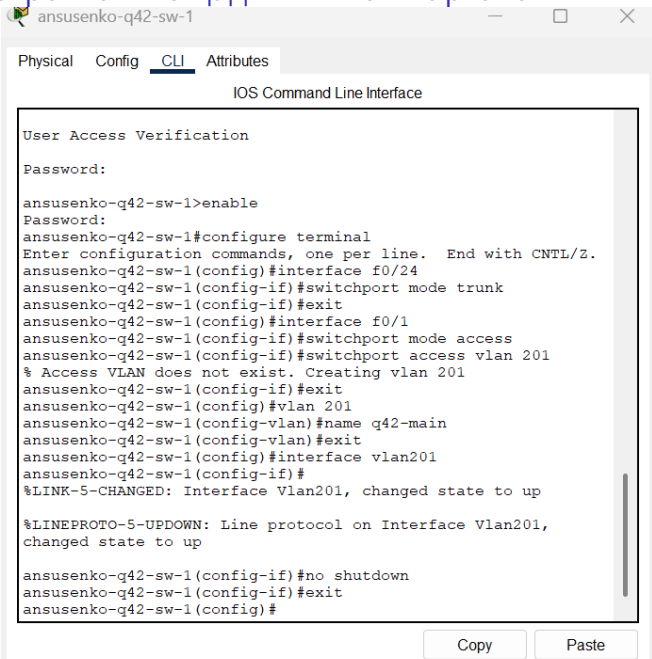
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

ansusenko-q42-gw-1(config-if)#exit
ansusenko-q42-gw-1(config)#interface f1/0.202
ansusenko-q42-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet1/0.202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up

ansusenko-q42-gw-1(config-subif)#encapsulation dot1Q 202
ansusenko-q42-gw-1(config-subif)#ip address 10.129.1.1 255.255.255.0
ansusenko-q42-gw-1(config-subif)#description q42-management
ansusenko-q42-gw-1(config-subif)#exit
ansusenko-q42-gw-1(config)#exit
ansusenko-q42-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
ansusenko-q42-gw-1#
```


Настройка площадки 42-го квартала



Настройка площадки 42-го квартала

ansusenko-hostel-gw-1

Physical Config CLI Attributes

IOS Command Line Interface

```
ansusenko-hostel-gw-1>enable
Password:
ansusenko-hostel-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ansusenko-hostel-gw-1(config)#interface g0/1
ansusenko-hostel-gw-1(config-if)#switchport trunk encapsulation dot1q
ansusenko-hostel-gw-1(config-if)#switchport mode trunk
ansusenko-hostel-gw-1(config-if)#exit
ansusenko-hostel-gw-1(config)#interface f0/1
ansusenko-hostel-gw-1(config-if)#switchport trunk encapsulation dot1q
ansusenko-hostel-gw-1(config-if)#switchport mode trunk

ansusenko-hostel-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
up

ansusenko-hostel-gw-1(config-if)#exit
ansusenko-hostel-gw-1(config)#vlan 202
ansusenko-hostel-gw-1(config-vlan)#name q42management
ansusenko-hostel-gw-1(config-vlan)#name q42management
ansusenko-hostel-gw-1(config-vlan)#name q42-management
ansusenko-hostel-gw-1(config-vlan)#exit
ansusenko-hostel-gw-1(config)#interface vlan202
ansusenko-hostel-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up

ansusenko-hostel-gw-1(config-if)#no shutdown
ansusenko-hostel-gw-1(config-if)#ip address 10.129.1.2 255.255.255.0
ansusenko-hostel-gw-1(config-if)#exit
ansusenko-hostel-gw-1(config)#vlan 301
ansusenko-hostel-gw-1(config-vlan)#vlan 301
ansusenko-hostel-gw-1(config-vlan)#name hostel-main
ansusenko-hostel-gw-1(config-vlan)#exit
ansusenko-hostel-gw-1(config)#interface vlan301
ansusenko-hostel-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up
```

Настройка площадки 42-го квартала

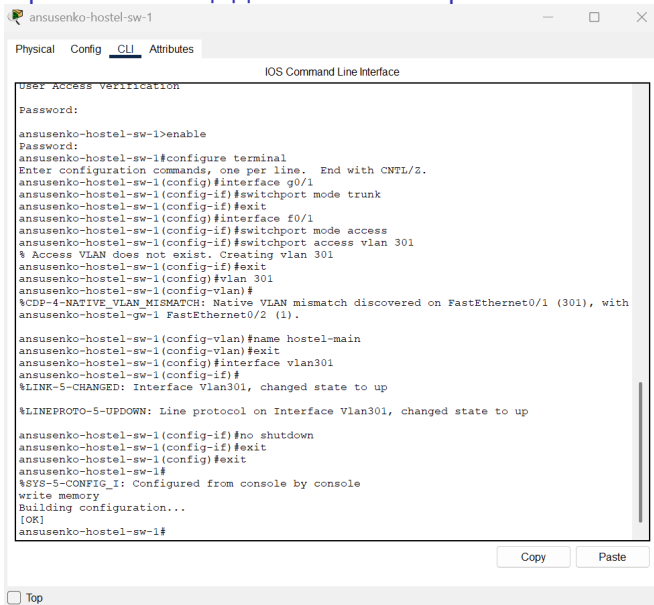
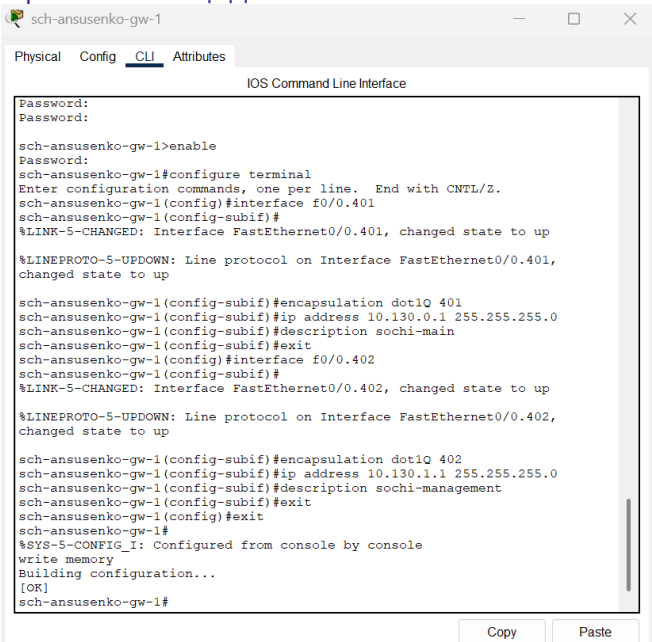


Рис.: Настройка интерфейсов коммутатора ansusenko-hostel-sw-1.

Настройка площадки в Сочи



The screenshot shows a network configuration window titled "sch-ansusenko-gw-1". It has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The window displays the "IOS Command Line Interface" with a series of commands and their outputs. The commands configure two interfaces, f0/0.401 and f0/0.402, with encapsulation, IP addresses, and descriptions. The interface f0/0.401 is configured with IP 10.130.0.1 and description "sochi-main". The interface f0/0.402 is configured with IP 10.130.1.1 and description "sochi-management". The configuration is saved to memory and the configuration is built.

```
sch-ansusenko-gw-1#enable
Password:
sch-ansusenko-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-ansusenko-gw-1(config)#interface f0/0.401
sch-ansusenko-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.401, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401,
changed state to up
sch-ansusenko-gw-1(config-subif)#encapsulation dot1Q 401
sch-ansusenko-gw-1(config-subif)#ip address 10.130.0.1 255.255.255.0
sch-ansusenko-gw-1(config-subif)#description sochi-main
sch-ansusenko-gw-1(config-subif)#exit
sch-ansusenko-gw-1(config)#interface f0/0.402
sch-ansusenko-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.402, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402,
changed state to up
sch-ansusenko-gw-1(config-subif)#encapsulation dot1Q 402
sch-ansusenko-gw-1(config-subif)#ip address 10.130.1.1 255.255.255.0
sch-ansusenko-gw-1(config-subif)#description sochi-management
sch-ansusenko-gw-1(config-subif)#exit
sch-ansusenko-gw-1(config)#exit
sch-ansusenko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
sch-ansusenko-gw-1#
```

Copy Paste

Настройка площадки в Сочи

 sch-ansusenko-sw-1

Physical Config CLI Attributes

IOS Command Line Interface

```

Password:

sch-ansusenko-sw-1>enable
Password:
sch-ansusenko-sw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
sch-ansusenko-sw-1(config)#interface f0/1
sch-ansusenko-sw-1(config-if)#switchport mode access
sch-ansusenko-sw-1(config-if)#switchport access vlan 401
% Access VLAN does not exist. Creating vlan 401
sch-ansusenko-sw-1(config-if)#exit
sch-ansusenko-sw-1(config)#vlan 401
sch-ansusenko-sw-1(config-vlan)#name sochi-main
sch-ansusenko-sw-1(config-vlan)#exit
sch-ansusenko-sw-1(config)#interface vlan401
sch-ansusenko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed
state to up

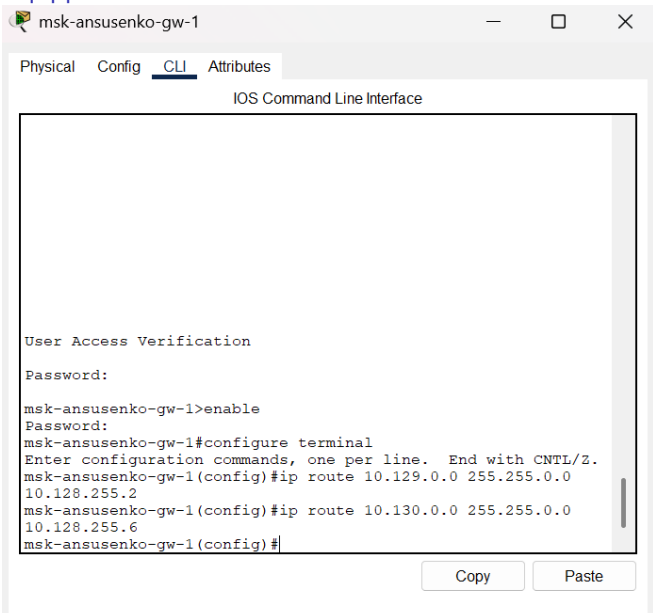
sch-ansusenko-sw-1(config-if)#no shutdown
sch-ansusenko-sw-1(config-if)#exit
sch-ansusenko-sw-1(config)#exit
sch-ansusenko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
sch-ansusenko-sw-1#

```

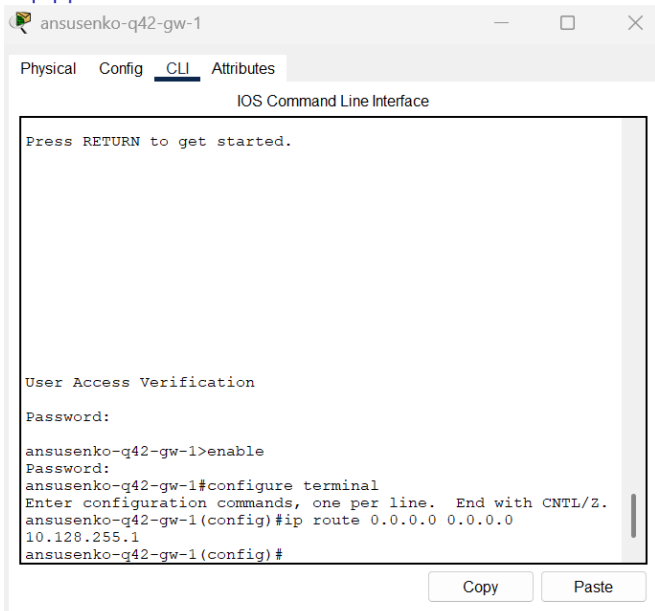
Copy Paste

☐ Top

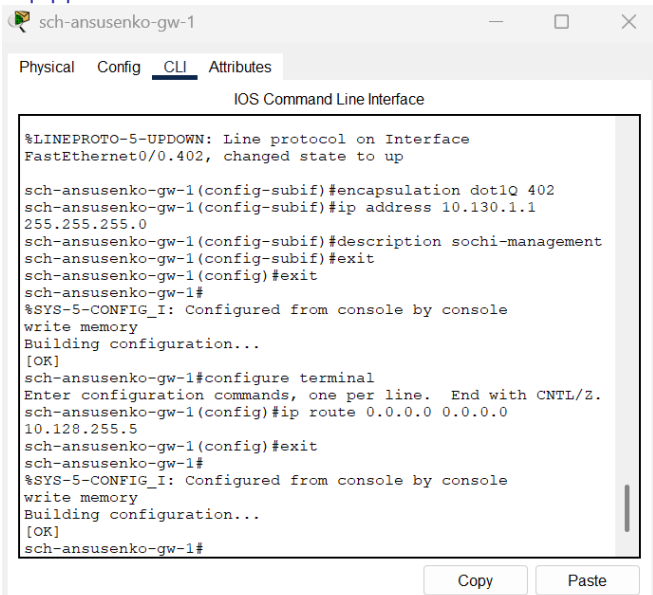
Настройка статической маршрутизации между площадками



Настройка статической маршрутизации между площадками



Настройка статической маршрутизации между площадками



The screenshot shows a network configuration window titled 'sch-ansusenko-gw-1'. It has four tabs: 'Physical', 'Config', 'CLI' (which is selected), and 'Attributes'. The main area is titled 'IOS Command Line Interface' and contains a terminal window with the following text:

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0.402, changed state to up

sch-ansusenko-gw-1(config-subif)#encapsulation dot1Q 402
sch-ansusenko-gw-1(config-subif)#ip address 10.130.1.1
255.255.255.0
sch-ansusenko-gw-1(config-subif)#description sochi-management
sch-ansusenko-gw-1(config-subif)#exit
sch-ansusenko-gw-1(config)#exit
sch-ansusenko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
sch-ansusenko-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-ansusenko-gw-1(config)#ip route 0.0.0.0 0.0.0.0
10.128.255.5
sch-ansusenko-gw-1(config)#exit
sch-ansusenko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
sch-ansusenko-gw-1#
```

At the bottom of the window, there are two buttons: 'Copy' and 'Paste'.

Настройка маршрутизации на 42 квартале

 ansusenko-q42-gw-1

Physical Config CLI Attributes

IOS Command Line Interface

```
Password:

ansusenko-q42-gw-1>enable
Password:
ansusenko-q42-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ansusenko-q42-gw-1(config)#ip route 0.0.0.0 0.0.0.0
10.128.255.1
ansusenko-q42-gw-1(config)#exit
ansusenko-q42-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
ansusenko-q42-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ansusenko-q42-gw-1(config)#ip route 10.129.128.0
255.255.128.0 10.129.1.2
ansusenko-q42-gw-1(config)#exit
ansusenko-q42-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
ansusenko-q42-gw-1#
```

Copy Paste

☐ Top

Настройка маршрутизации на 42 квартале

 ansusenko-hostel-gw-1

Physical Config CLI Attributes

IOS Command Line Interface

```
FastEthernet0/1 (301).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered
on FastEthernet0/2 (1), with ansusenko-hostel-sw-1
FastEthernet0/1 (301).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered
on FastEthernet0/2 (1), with ansusenko-hostel-sw-1
FastEthernet0/1 (301).

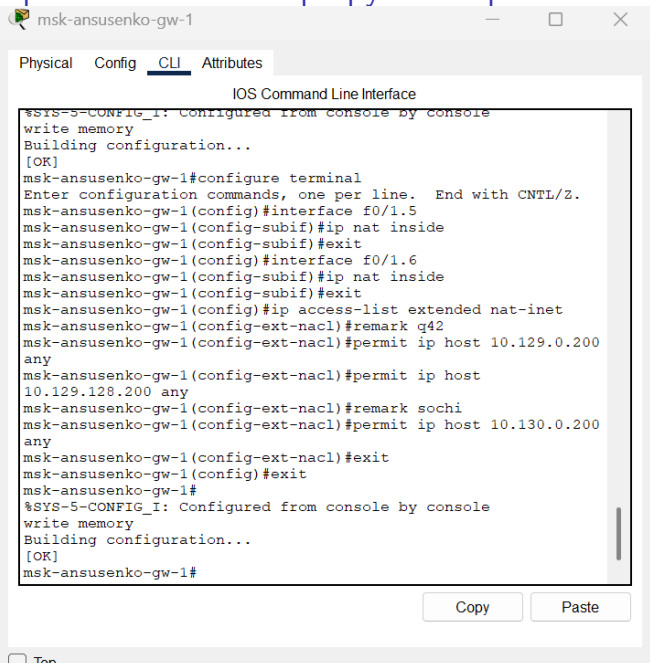
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered
on FastEthernet0/2 (1), with ansusenko-hostel-sw-1
FastEthernet0/1 (301).

ansusenko-hostel-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ansusenko-hostel-gw-1(config)#ip routing
ansusenko-hostel-gw-1(config)#ip route 0.0.0.0 0.0.0.0
10.129.1.1
ansusenko-hostel-gw-1(config)#exit
ansusenko-hostel-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
ansusenko-hostel-gw-1#
```

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Настройка NAT на маршрутизаторе msk-ansusenko-gw-1



The screenshot shows a terminal window titled "msk-ansusenko-gw-1" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The terminal output shows the configuration of NAT on the router. It starts with a system message, followed by "write memory" and "Building configuration...". The configuration commands include setting the terminal, configuring interfaces f0/1.5 and f0/1.6 with "ip nat inside", creating an extended access list "nat-inet" with two "permit" rules for hosts 10.129.0.200 and 10.130.0.200, and finally exiting the configuration mode. The terminal ends with another system message and "Building configuration...".

```
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
msk-ansusenko-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-ansusenko-gw-1(config)#interface f0/1.5
msk-ansusenko-gw-1(config-subif)#ip nat inside
msk-ansusenko-gw-1(config-subif)#exit
msk-ansusenko-gw-1(config)#interface f0/1.6
msk-ansusenko-gw-1(config-subif)#ip nat inside
msk-ansusenko-gw-1(config-subif)#exit
msk-ansusenko-gw-1(config)#ip access-list extended nat-inet
msk-ansusenko-gw-1(config-ext-nacl)#remark q42
msk-ansusenko-gw-1(config-ext-nacl)#permit ip host 10.129.0.200
any
msk-ansusenko-gw-1(config-ext-nacl)#permit ip host
10.129.128.200 any
msk-ansusenko-gw-1(config-ext-nacl)#remark sochi
msk-ansusenko-gw-1(config-ext-nacl)#permit ip host 10.130.0.200
any
msk-ansusenko-gw-1(config-ext-nacl)#exit
msk-ansusenko-gw-1(config)#exit
msk-ansusenko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
msk-ansusenko-gw-1#
```

Buttons: Copy Paste

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Проверка корректности настроек

ansusenko-q42-sw-1

Physical **Config** CLI Attributes

INTERFACE
FastEthernet0/1
FastEthernet0/2
FastEthernet0/3
FastEthernet0/4
FastEthernet0/5
FastEthernet0/6
FastEthernet0/7
FastEthernet0/8
FastEthernet0/9
FastEthernet0/10
FastEthernet0/11
FastEthernet0/12
FastEthernet0/13
FastEthernet0/14
FastEthernet0/15

FastEthernet0/1

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

Access VLAN

Tx Ring Limit

Equivalent IOS Commands

```
ansusenko-q42-sw-1(config-if)#exit
ansusenko-q42-sw-1(config)#interface FastEthernet0/2
ansusenko-q42-sw-1(config-if)#
ansusenko-q42-sw-1(config-if)#exit
ansusenko-q42-sw-1(config)#interface FastEthernet0/1
ansusenko-q42-sw-1(config-if)#
```

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Проверка корректности настроек

ansusenko-hostel-sw-1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings
- SWITCHING**
- VLAN Database
- INTERFACE**
- FastEthernet0/1
- FastEthernet0/2
- FastEthernet0/3
- FastEthernet0/4
- FastEthernet0/5
- FastEthernet0/6
- FastEthernet0/7
- FastEthernet0/8
- FastEthernet0/9
- FastEthernet0/10

FastEthernet0/1

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

Access VLAN

Tx Ring Limit

Equivalent IOS Commands

```
ansusenko-hostel-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ansusenko-hostel-sw-1(config)#interface FastEthernet0/1
ansusenko-hostel-sw-1(config-if)#
```

Проверка корректности настроек

sch-ansusenko-sw-1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

SWITCHING

- VLAN Database

INTERFACE

- FastEthernet0/1**
- FastEthernet0/2
- FastEthernet0/3
- FastEthernet0/4
- FastEthernet0/5
- FastEthernet0/6
- FastEthernet0/7
- FastEthernet0/8
- FastEthernet0/9
- FastEthernet0/10

FastEthernet0/1

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☒ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

Access VLAN

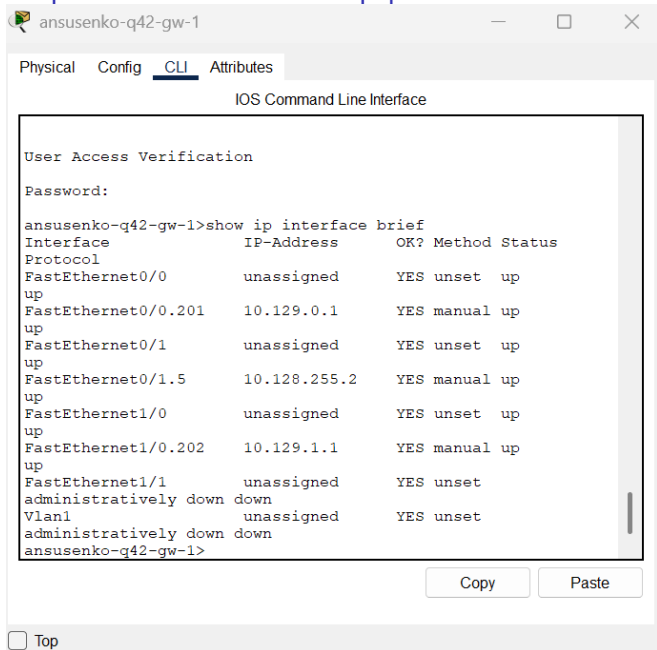
Tx Ring Limit

Equivalent IOS Commands

```
[OK]
sch-ansusenko-sw-1#sch-ansusenko-sw-1#
sch-ansusenko-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-ansusenko-sw-1(config)#interface FastEthernet0/1
sch-ansusenko-sw-1(config-if)#
```

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Проверка состояния интерфейсов



The screenshot shows a web-based CLI interface for a device named 'ansusenko-q42-gw-1'. The 'CLI' tab is selected, displaying the 'IOS Command Line Interface'. The prompt is 'ansusenko-q42-gw-1>'. The command 'show ip interface brief' has been entered, resulting in a table of interface statuses. Below the table, there are 'Copy' and 'Paste' buttons, and a 'Top' link at the bottom left.

Physical Config CLI Attributes

IOS Command Line Interface

User Access Verification

Password:

```
ansusenko-q42-gw-1>show ip interface brief
```

Interface	IP-Address	OK?	Method	Status
Protocol				
FastEthernet0/0	unassigned	YES	unset	up
FastEthernet0/0.201	10.129.0.1	YES	manual	up
FastEthernet0/1	unassigned	YES	unset	up
FastEthernet0/1.5	10.128.255.2	YES	manual	up
FastEthernet1/0	unassigned	YES	unset	up
FastEthernet1/0.202	10.129.1.1	YES	manual	up
FastEthernet1/1	unassigned	YES	unset	administratively down
Vlan1	unassigned	YES	unset	administratively down

ansusenko-q42-gw-1>

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Проверка состояния интерфейсов

 sch-ansusenko-gw-1

Physical Config CLI Attributes

IOS Command Line Interface

User Access Verification

Password:

```
sch-ansusenko-gw-1>show ip interface brief
```

Interface	IP-Address	OK?	Method	Status
Protocol				
FastEthernet0/0	unassigned	YES	unset	up
FastEthernet0/0.6	10.128.255.6	YES	manual	up
FastEthernet0/0.401	10.130.0.1	YES	manual	up
FastEthernet0/0.402	10.130.1.1	YES	manual	up
FastEthernet0/1	unassigned	YES	unset	administratively down
Vlan1	unassigned	YES	unset	administratively down

```
sch-ansusenko-gw-1>
```

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Проверка VLAN

ansusenko-sw-1

Physical Config CLI Attributes

IOS Command Line Interface

```
ansusenko-sw-1(config-if)#
ansusenko-sw-1(config-if)#exit
ansusenko-sw-1(config)#interface GigabitEthernet0/1
ansusenko-sw-1(config-if)#ansusenko-sw-1(config-if)#exit
ansusenko-sw-1(config)#exit
ansusenko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

ansusenko-sw-1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
4	nat	active	
5	q42	active	
6	sochi	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
ansusenko-sw-1#
```

Проверка VLAN

 sch-ansusenko-sw-1

Physical Config CLI Attributes

IOS Command Line Interface

```
%SYS-5-CONFIG_I: Configured from console by console  
sch-ansusenko-sw-1#show vlan brief
```

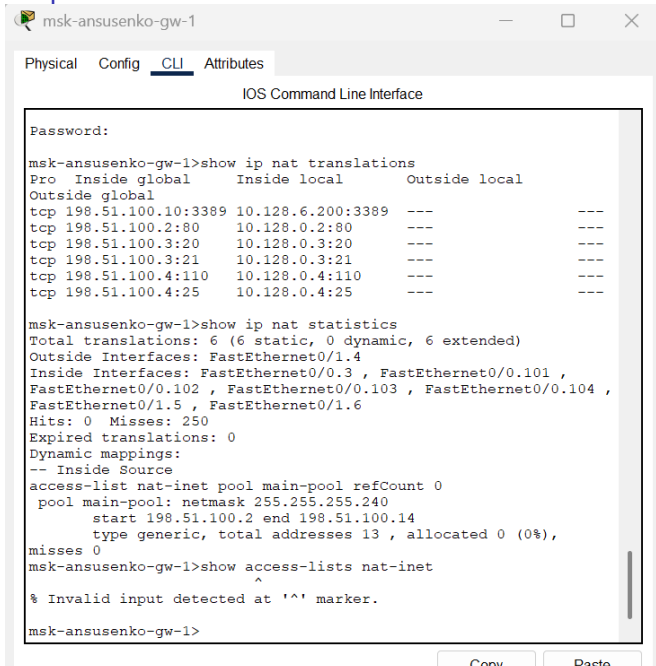
VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/12, Fa0/13, Fa0/14 Fa0/16, Fa0/17, Fa0/18 Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24
6	sochi	active	
401	sochi-main	active	Fa0/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
sch-ansusenko-sw-1#
```

Copy Paste

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Проверка NAT



The screenshot shows a network device's CLI interface. At the top, there are tabs for 'Physical', 'Config', 'CLI' (which is selected), and 'Attributes'. Below the tabs is the title 'IOS Command Line Interface'. The main area displays the output of several commands entered in the CLI.

```
msk-ansusenko-gw-1>show ip nat translations
Pro  Inside global      Inside local      Outside local
Outside global
tcp 198.51.100.10:3389 10.128.6.200:3389 ---
tcp 198.51.100.2:80    10.128.0.2:80    ---
tcp 198.51.100.3:20    10.128.0.3:20    ---
tcp 198.51.100.3:21    10.128.0.3:21    ---
tcp 198.51.100.4:110   10.128.0.4:110   ---
tcp 198.51.100.4:25    10.128.0.4:25    ---

msk-ansusenko-gw-1>show ip nat statistics
Total translations: 6 (6 static, 0 dynamic, 6 extended)
Outside Interfaces: FastEthernet0/1.4
Inside Interfaces: FastEthernet0/0.3 , FastEthernet0/0.101 ,
FastEthernet0/0.102 , FastEthernet0/0.103 , FastEthernet0/0.104 ,
FastEthernet0/1.5 , FastEthernet0/1.6
Hits: 0 Misses: 250
Expired translations: 0
Dynamic mappings:
-- Inside Source
access-list nat-inet pool main-pool refCount 0
pool main-pool: netmask 255.255.255.240
start 198.51.100.2 end 198.51.100.14
type generic, total addresses 13 , allocated 0 (0%),
misses 0
msk-ansusenko-gw-1>show access-lists nat-inet
^
% Invalid input detected at '^' marker.

msk-ansusenko-gw-1>
```

Ответы на контрольные вопросы

1. **Пример настройки статической маршрутизации между двумя подсетями организации:**

```
ip route 192.168.2.0 255.255.255.0 192.168.1.2 ip route  
192.168.3.0 255.255.255.0 192.168.1.2
```

Ответы на контрольные вопросы

2. Процесс обращения устройства из одного VLAN к устройству из другого VLAN:

- ▶ Устройство-источник отправляет ARP-запрос на шлюз по умолчанию
- ▶ Пакет передается на маршрутизатор (или L3-коммутатор)
- ▶ Маршрутизатор проверяет таблицу маршрутизации и определяет исходящий интерфейс
- ▶ Маршрутизатор отправляет ARP-запрос в целевом VLAN
- ▶ После получения MAC-адреса пакет инкапсулируется и отправляется устройству-назначения

Ответы на контрольные вопросы

3. Как проверить работоспособность маршрута?

```
ping <IP-адрес назначения> traceroute <IP-адрес назначения>  
show ip route <IP-адрес>
```

Ответы на контрольные вопросы

4. Как посмотреть таблицу маршрутизации?

`show ip route` полная таблица `show ip route static` только статические маршруты `show ip route connected` только подключенные сети

Выводы

- ▶ В ходе выполнения лабораторной работы настроена статическая маршрутизация между тремя площадками (провайдер, 42-й квартал в Москве, филиал в Сочи) с использованием технологии VLAN и subinterface-ов. На маршрутизаторах настроены статические маршруты, включая маршруты по умолчанию и суммарные маршруты. На маршрутизаторе msk-ansusenko-gw-1 выполнена настройка NAT для трансляции адресов внутренних хостов.